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# LiMOB, an Unsymmetrical Nonaromatic Orthoborate Salt for Nonaqueous Solution Electrochemical Applications

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Synthesis, characterization, ionic conductivity, electrochemical stability, and lithium-ion transport number of an asymmetric version of the successful orthoborate lithium salt lithium bis(oxalato)borate (LiBOB), are described. The salt, lithium (malonato oxalato) borate (LiMOB), is thermally stable up to 273°C. It is poorly soluble in common organic solvents such as 1,2-dimethoxyethane, tetrahydrofuran, acetonitrile, dimethyl carbonate, and propylene carbonate, but has moderate solubility in γ-butyrolactone (GBL) and N,N-dimethylformamide (DMF), and high solubility in dimethyl sulfoxide (DMSO). The 0.5 M solutions of LiMOB in GBL, DMSO, and DMF show high conductivities ($5.0 \times 10^{-3}$ S cm$^{-1}$, $5.1 \times 10^{-3}$ S cm$^{-1}$, and $11.8 \times 10^{-3}$ S cm$^{-1}$ at 25°C, respectively) relative to most 0.5 M nonaqueous solutions. Even the room temperature conductivity of the 0.08 M LiMOB-PC solution is close to $10^{-3}$ S cm$^{-1}$. The conductivities of LiMOB solutions are close to those of LiBOB solutions, which has been demonstrated in recent studies to be a successful substitute for $LiPF_6$ in lithium-ion batteries. The cyclic voltammograms show that LiMOB solutions in PC and GBL have electrochemical stability as high as 4.3 V *vs.* $Li^+$/Li. The lithium-ion transport number at room temperature for 0.5 M LiMOB in DMSO-$d_6$ was 0.36, obtained from self-diffusivity measurement by the pulsed field gradient spin echo nuclear magnetic resonance technique. The highest occupied molecular orbital energy, which is usually correlated with electrochemical stability has been calculated, and electrostatic potential maps for the new anion and its fluorinated derivatives are presented. They suggest that the fluorinated analogue of LiMOB should be more stable and more soluble, and may have unique properties.
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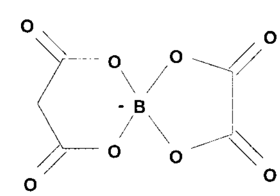
While lithium salts play a vital role in electrolytes for many electrochemical devices, there are many problems in their applications. The problems encountered with the commonly used lithium salts in primary and secondary lithium batteries have stimulated the search for substitutes.[1] In recent years, a new class of lithium salts with chelated orthoborate anions was initiated by Barthel *et al.*[2-13] Lithium chelatoborate salts are generally thermally stable and have good ionic conductivites and wide electrochemical windows in electrolytic solutions. This is largely due to the extensive charge delocalization in their anions, consequent on the strongly electron-withdrawing groups attached to the boron-containing charge centers.

In our recent publications,[13,14] we have described the synthesis and properties of new lithium salts, which have fluorine-free and aromatic-free orthoborate anions, namely, lithium bis(oxalato)borate (LiBOB) and lithium bis(malonato)borate (LiBMB). The former was also disclosed by Lischka *et al.* in Germany.[15] These salts are quite thermally stable and have anions containing eight oxygen atoms, four of which are carbonyl oxygens. Because there is only a single negative charge distributed across them, these anions are very weakly coordinating. In consequence, their solutions in molecular solvents are extremely conductive and the electrochemical stability window is also high, 4.5 V *vs.* $Li^+$/Li.

Continued research on LiBOB solutions in cell applications[16,17] at the Army Research Laboratory has shown that the compound is able to form a very effective passivation film on Al, and superior protective films on both carbonaceous anodes and metal oxide cathodes. This property permits extraordinary cell performance at elevated temperatures relative to the industry standard, $LiPF_6$. Even more surprising, LiBOB can effectively stabilize graphite materials in propylene carbonate (PC)-rich or even neat PC solutions, a property unknown for other commonly used lithium salts.[17] Thus LiBOB provides a good substitute for $LiPF_6$ in the lithium-ion battery industry, one that combines performance with economy and non-toxicity.

In this paper we present data on a variation on the earlier theme in which the anion is asymmetric, combining $BOB^-$ and $BMB^-$ parts to give lithium (malonato oxalato) borate (LiMOB). The anion structure is shown below (Scheme I). The synthesis, characterization, ionic conductivities, electrochemical stability, and lithium-ion transport number of LiMOB are described.



**Scheme I.**

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## Experimental

*Synthesis.*—LiMOB was synthesized according to Scheme II, and followed the procedures described previously.[13] 0.025 mol lithium tetramethanolatoborate [$LiB(OCH_3)_4$], 0.025 mol di(trimethylsilyl) malonate (DTMSM), and 0.025 mol di(trimethylsilyl) oxalate (DTMSO) in 150 mL anhydrous acetontrile (MeCN) were stirred in an oil bath of 45-50°C overnight. The white solid $LiB(OCH_3)_4$ disappeared and the solution became turbid indicating that a new solid formed. The new solid, also white, was filtered and washed twice with fresh anhydrous MeCN. Then it was dissolved in a small amount of anhydrous N,N-dimethylformamide (DMF). After filtration, most of the solvent DMF in the filtrate was evaporated on a rotary evaporator at reduced pressure. The residue was poured into a large amount of anhydrous diethyl ether to yield a slightly yellowish white solid, which was filtered and dried in a vacuum oven at 80°C for 48 h, with a yield of *ca.* 85%. Mass spectroscopy showed no signals for the starting material, only some decomposition products with masses ($m/z$) for CO, $O_2$, and $CO_2$. Nuclear magnetic resonance (NMR) spectra in dimethyl sulfoxide (DMSO)-$d_6$: $^1$H, δ 3.49 ppm (s); $^{13}$C, δ 165.76, 158.66, and 38.44 ppm; $^7$Li (reference to 9.7 m LiCl in $D_2O$), δ −0.13 ppm; and $^{11}$B (reference to $BF_3 \cdot Et_2O$ in $CDCl_3$), δ 10.11 ppm. Elemental analysis: found C 28.28%, H 0.86%; calculated for LiMOB ($LiBC_5H_2O_8$) C 28.85%, H 0.96%.

$$LiB(OCH_3)_4 + (CH_3)_3SiOOCCH_2COOSi(CH_3)_3 + (CH_3)_3SiOOCCOOSi(CH_3)_3 \xrightarrow{CH_3CN} LiB[(OCO)_2CH_2][(OCO)_2] + 4\ CH_3OSi(CH_3)_3$$

**Scheme II.** Synthesis of LiMOB.

*Measurements*.—Mass spectroscopy (EI, source temperature 250°C) was carried out with a Finnigan MAT model 312 using ether as carrier. $^1H$ and $^{13}C$ NMR spectra, using tetramethyl silane as internal reference, were obtained on a Varian Gemini 300, while $^7Li$ and $^{11}B$ NMR spectra were collected using a Varian Inova 400, with 9.7 m LiCl in $D_2O$ and $BF_3 \cdot Et_2O$ in $CDCl_3$ as external references, respectively.

Differential thermal analysis-thermogravimetric analysis (DTA-TGA) measurement of LiMOB was conducted in a Setaram TG-DTA 92. The sample was contained in an alumina crucible and was kept at 20°C for 1 h, then heated to 800°C with a ramp of 2°C per min and finally cooled to room temperature at a rate of 10°C per min in a helium atmosphere. The thermal properties of LiMOB solutions in DMSO were obtained using a simple DTA instrument, which was reported previously.[18]

The ac impedance conductivities were measured using a Hewlett Packard 4192A LF impedance analyzer in a frequency range from 5 Hz to 13 MHz, with homemade dip-type cells containing two paralleled platinum discs. The cell constants ranged from 0.7 to 1.3 $cm^{-1}$, and were calibrated by a 0.1 m KCl aqueous solution.

The electrochemical properties were measured by cyclic voltammetric methods on an EG&G potentiostat/galvanostat model 273, using a three-electrode dip cell with platinum wire as working electrode and lithium as counter and reference electrodes.

Self-diffusivity data were obtained using NMR, with a pulsed field gradient spin echo probe operating at 300 MHz for protons in a Varian Infinity 300 Spectrometer. Our application of this technique was described in Ref. 19 and 14.

*Computations*.—B3LYP/6-31 + G** and B3LYP/6-311 + G** calculations were used to construct anion models as described previously.[14] Highest occupied molecular orbital (HOMO) energies were calculated at the B3LYP/6-31 + G** geometries using B3LYP/6-31 + G** and HF/6-31 + G** calculations.

## Results and Discussion

*Physical and chemical properties*.—The DTA-TGA thermogram shows LiMOB cannot be fused before it decomposes at 273°C. This temperature is 30°C lower than for LiBOB (302°C) but is higher than for LiBMB (245°C).[13]

Like LiBOB and LiBMB, LiMOB is stable in organic electrolytic solutions but it decomposes by hydrolysis and alcoholysis, yielding benign products. Unfortunately, it is not very soluble. Tests show that LiMOB is only slightly soluble in many common solvents, such as 1,2-dimethylethane (DME), tetrahydrofuran (THF), MeCN, dimethyl carbonate (DMC), and dissolves only to 0.1 M PC at room temperature. However, it is moderately soluble in γ-butyrolactone (GBL, 0.7 M at 25°C), DMF, ethyl methyl sulfone (EMS), and its eutectic with dimethyl sulfone (DMS), and has high solubility in DMSO (1.5 M at 25°C).

The -$CH_2$- group between the two carbonyls in LiMOB reduces the distribution of the negative charge over the whole molecule, relative to LiBOB, which is probably the reason that LiMOB has a lower solubility in polar solvents than LiBOB. It will therefore be interesting to replace the protons of LiMOB with electron withdrawing groups such as fluorine or fluorinated alkyl groups. The fluorinated analogs should be more soluble, and have other useful properties.

*Conductivity*.—Figure 1 shows the temperature dependence of ionic conductivity of LiMOB solutions in different neat solvents such as DMF, DMSO, GBL, and PC. We also include data for solutions in the eutectic mixture of EMS-DMS (85:15 molar ratio) which has the largest electrochemical window of any solvent system.[20] The 0.5 M solutions of LiMOB in DMF and DMSO show extremely high conductivities of $11.8 \times 10^{-3}$ S $cm^{-1}$ and $5.1 \times 10^{-3}$ S $cm^{-1}$ at 25°C, respectively. The room temperature conductivity of 0.47 M LiMOB-GBL solution is also high, $5.0 \times 10^{-3}$ S $cm^{-1}$. Even the 0.08 M LiMOB in PC solution shows room temperature conductivity as high as $9.3 \times 10^{-4}$ S $cm^{-1}$, indicating that the ions in LiMOB solutions are highly mobile.

Figure 2 compares the conductivities of LiMOB solutions in different solvents with those of LiBOB, LiBMB, and the other commonly studied lithium electrolyte salt, the bis(trifluoromethanesulfonyl)imide LiTFSI. It is seen that LiMOB solutions show conductivities between those of LiBOB and LiBMB solutions.

The variations of the isothermal conductivities of LiMOB in DMSO and GBL solutions with salt concentration are shown in Fig. 3 and 4, respectively. When the temperature is above 25°C, the conductivities of LiMOB-DMSO solutions are almost independent of concentration from 0.7 to 1.5 M, which could be advantageous. This behavior for LiMOB-GBL solutions from 0.4 to 0.7 M is maintained even when the temperature is −25°C. At −25°C, the conductivities are still higher than $10^{-3}$ S $cm^{-1}$, indicating the potential applications of GBL solutions at very low temperatures.

In Fig. 3 is also noted the temperature $T_c$ at which the solvate (*i.e.*, the complex of solvent and salt) crystals formed during cooling in LiMOB-DMSO solutions from the DTA measurements. It is seen that the solvate crystals melt at around 12.8°C for 0.2 M, 4.8°C for

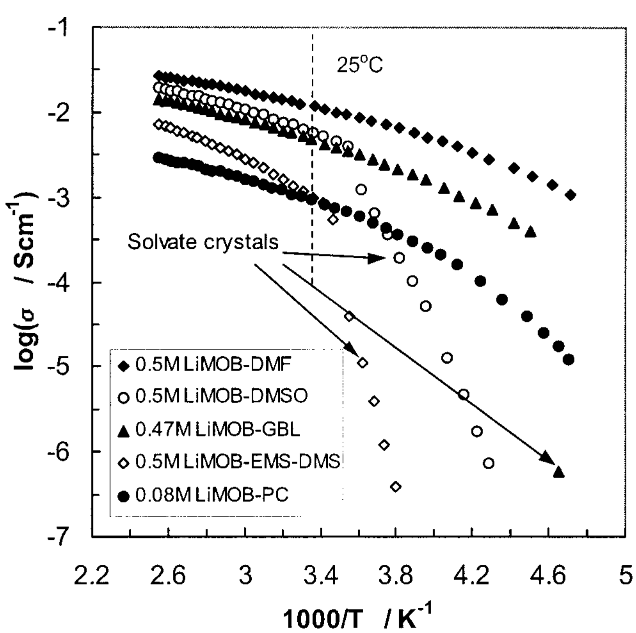


**Figure 1.** Temperature dependence of ionic conductivities of LiMOB solutions in different solvents.

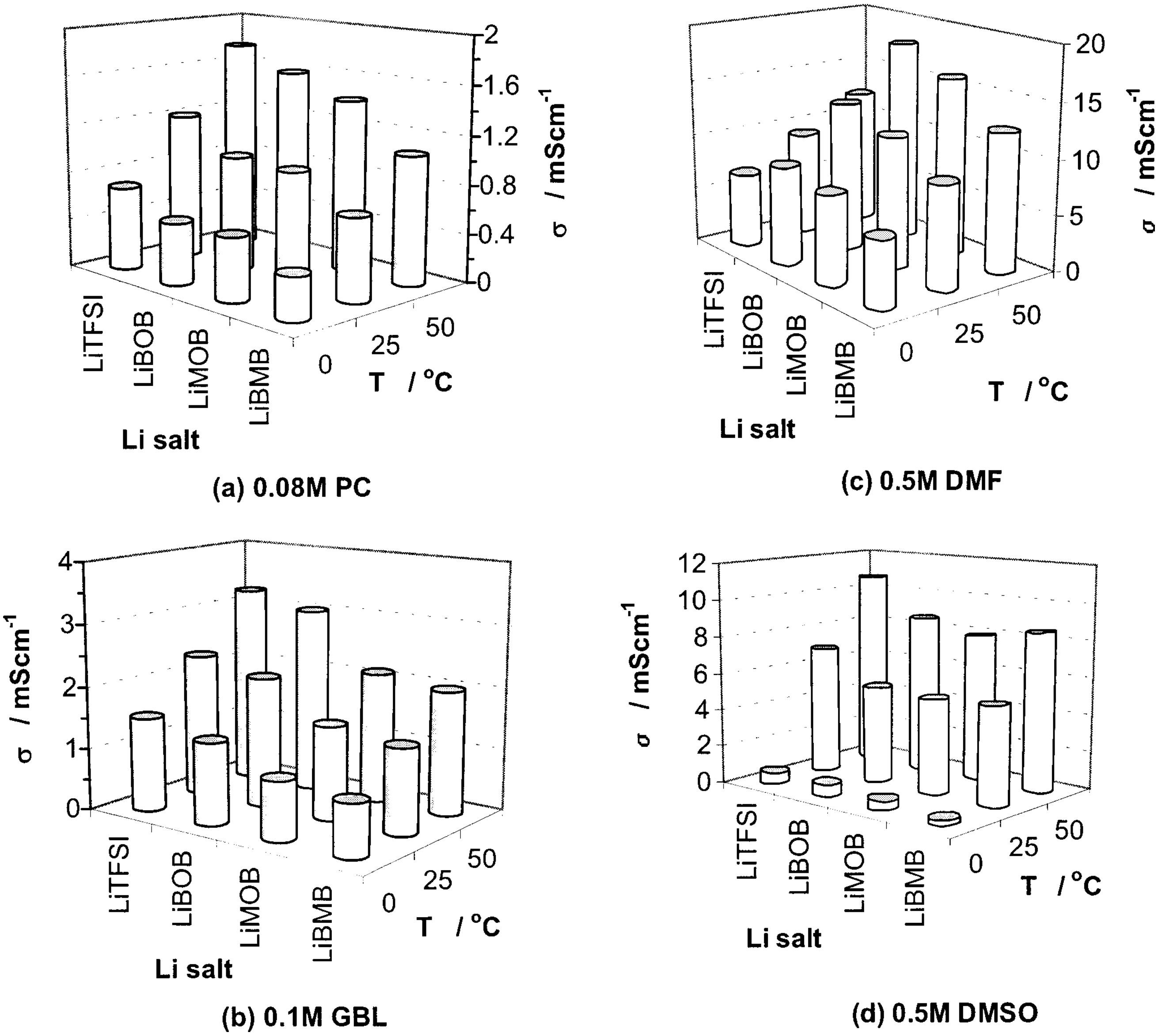


**Figure 2.** Conductivity comparison of LiMOB solutions in different solvents with those of LiBOB, LiBMB, and LiTFSI in the solvent.

0.5 M, 0°C for 0.7 M, −7.9°C for 1.0 M, −9.7°C for 1.3 M, and −13.8°C for 1.5 M solutions. The conductivity of the 1.5 M solution shows sudden decrease at a temperature of *ca.* 11°C below the temperature at which the solvate crystal melts, indicating that this solution supercools easily.

*Self-diffusivity and lithium-ion transport number*.—Stejskal-Tanner plots for lithium (from $^7$Li) and the anion species (from $^1$H) for 0.5 M LiMOB in DMSO-$d_6$ at 25°C are shown in Fig. 5. The linearity in these semilogarithmic plots over more than one order of magnitude in echo amplitude demonstrates the homogeneity of the pulses. Self-diffusivities for lithium and hydrogen-containing species are obtained from the slopes of the Stejskal-Tanner plots, as described elsewhere.[14,19] The values at 25°C are $1.64 \times 10^{-10}$ and $2.32 \times 10^{-10}$ $m^2$ $s^{-1}$, respectively. It is seen that lithium diffusivity is slightly smaller than the anion diffusivity, meaning that the conduction is still mainly anionic.

For this case the calculated conductivity from Nernst-Einstein equation exceeds the measured value by 58%, which is a larger deviation than for the symmetric anion $BMB^-$ ($\sigma_{NE}/\sigma_{exp} = 1.50$ in DMSO-$d_6$) and the "giant anion" salt, LiBPFPB ($\sigma_{NE}/\sigma_{exp} = 1.26$ in DMSO).[14] However it is still small relative to values

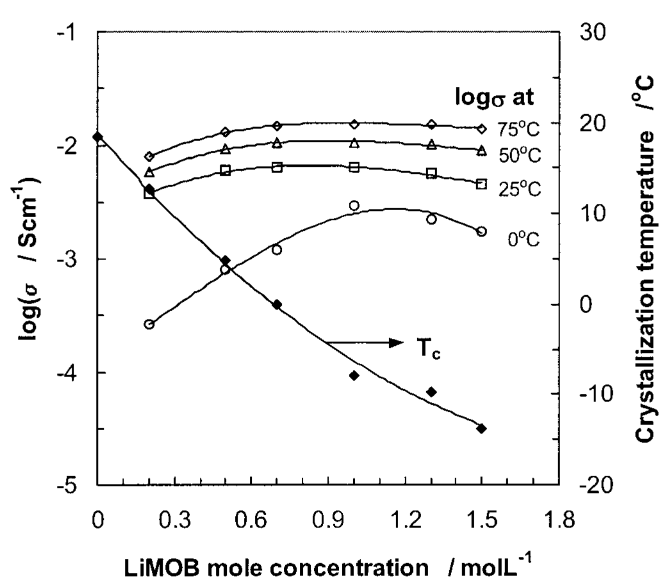


**Figure 3.** Variations of isothermal conductivities and crystallization temperatures of LiMOB-DMSO solutions with salt concentration.

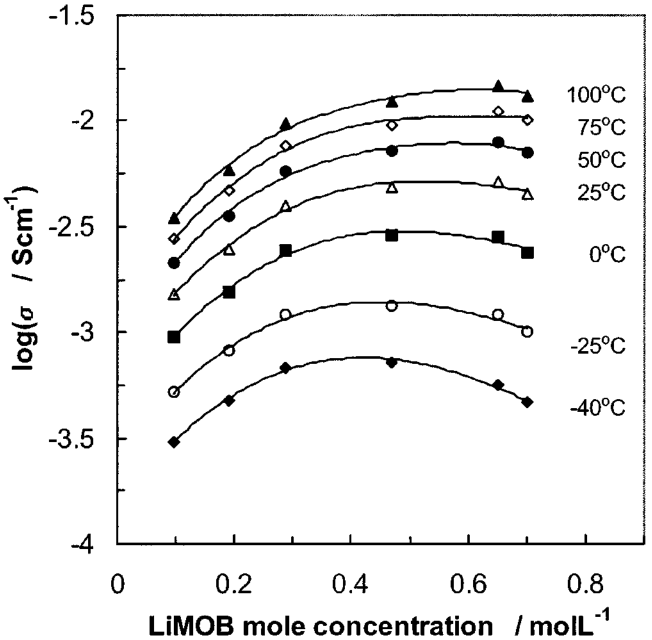


**Figure 4.** Variation of isothermal conductivities of LiMOB-GBL solutions with salt concentration.

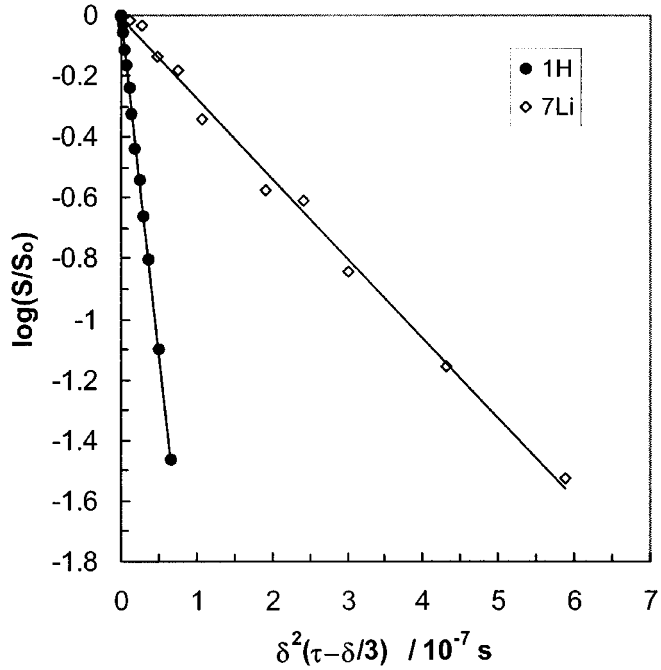


**Figure 5.** Stejskal-Tanner plots for $^7$Li and anion ($^1$H) for 0.5 M LiMOB in DMSO-$d_6$. Note strict linearity over more than one order of magnitude in log($S/S_0$).

encountered for more common salts in moderate dielectric constant solvents ($\epsilon = 47.2$ at 25°C for normal DMSO). This finding confirms the highly dissociated nature of LiMOB in moderate dielectric constant solvents. Supporting this conclusion is the similarity of the $\sigma_{Li,NE}/\sigma_{exp}$ values in DMSO-$d_6$ (column 5 in Table I) for LiMOB and LiBMB, 0.65 and 0.64,[14] respectively.

For the lithium-ion transport number $t_{Li+}$, we obtain, using the definition $t_{Li+} = D_{Li+}/(D_{Li+} + D_{MOB-})$, the value of 0.41. Making an exaggerated[14] correction for ion pairing, we obtain $t_{Li+} = 0.36$. Both values are slightly smaller than found in the symmetric bis(malonato)borate anion case and also the perfluorinated giant anion case in nondeuterated DMSO, $t_{Li+} = 0.55$.[14]

*Electrochemical stability*.—The electrochemical stabilities of LiMOB solutions in PC and GBL on platinum electrodes are shown in Fig. 6. The electrochemical oxidation potential can be obtained as *ca.* 4.3 and 4.4 V *vs.* $Li^+/Li$ for the solutions of LiMOB in 0.08 M PC and 0.7 M GBL, with the cutoff criteria of current density to be 0.1 mA $cm^{-2}$. This is slightly lower than the 4.5 V stability for LiBOB-PC solutions,[13] but is higher than for the LiBMB-PC solution (*ca.* 4.0 V). Solutions of salts containing a $-CH_2-$ group between the two C = O groups (such as LiMOB and LiBMB) have lower electrochemical stabilities than solutions of LiBOB. This is probably due to the acidity of the protons in the $-CH_2-$ group caused by the strong electron-withdrawing effect of the C = O groups. The electrochemical stability of LiMOB and LiBMB should be improved by replacing the acidified protons with other groups, such as F or $CF_3$, as discussed below.

Figure 7 compares the first and third cyclic voltammograms (CVs) of LiMOB solutions in PC and GBL on Pt electrodes, respectively, where the potential range was limited from 2.2 V to 0 V *vs.* $Li^+/Li$. These peaks are similar to those of nonaqueous Li salt solutions on platinum electrodes,[21,22] with relevant processes being the following: bulk Li deposition below 0 V, bulk Li stripping at *ca.* 0.75 V, Li underpotential deposition (UPD) at *ca.* 0.65-0.70 V and 0.38 V (two cathodic peaks), dissolution of UPD Li at 0.95 V and 1.40-1.45 V (two anodic peaks), reduction of residual water at *ca.* 1.70 V, and a corresponding anodic peak at *ca.* 2.0-2.1 V which is probably related to the oxidation of the formed hydrogen by the

**Table I. Self-diffusivities, lithium-ion transport number, and degree of dissociation of Li salts in 0.5 M solutions at 25°C.**

| 0.5 M solution | $D_{Li}$ ($m^2\ s^{-1}$) | $D_{anion}$ ($m^2\ s^{-1}$) | $\sigma_{exp}$ (mS $cm^{-1}$) | $\sigma_{Li,NE}/\sigma_{exp}$ [a] | $\sigma_{NE}/\sigma_{exp}$ [a] | $\xi$ [b] ($m^2\ s^{-1}$) | $D_{Li}/(D_{Li} + D_{anion})$ | $t_{Li+}$ [c] | $\alpha$ [d] |
|---|---|---|---|---|---|---|---|---|---|
| LiMOB-DMSO-$d_6$ | $1.64 \times 10^{-10}$ | $2.32 \times 10^{-10}$ | 4.71 | 0.65 | 1.58 | $2.51 \times 10^{-10}$ | 0.41 | 0.36 | 0.7 |
| LiBOB-DMSO | $2.12 \times 10^{-10}$ | — | 5.24 | 0.76 | — | — | — | — | — |
| LiBMB DMSO | $1.74 \times 10^{-10}$ | — | 4.99 | 0.66 | — | — | — | — | — |
| DMSO-$d_6$ | $1.52 \times 10^{-10}$ | $2.04 \times 10^{-10}$ | 4.45 | 0.64 | 1.50 | $2.37 \times 10^{-10}$ | 0.43 | 0.39 | 0.7 |

[a] $\sigma_{Li,NE} = cF^2(D_{Li+})/RT$, and $\sigma_{NE} = cF^2(D_{Li+} + D_{anion})/RT$
[b] $\xi = RT\sigma_{exp}/(cF^2)$
[c] $t_{Li+} = (D_{Li} - D_{anion} + \xi)/2\xi$
[d] $\alpha = (\xi - D_{Li})/2D_{anion} + 0.5$. The value of $\alpha$ is a lower bound because it is calculated by assuming that the Nernst-Einstein equation is an exact relation for concentrated ionic systems, which is certainly incorrect. Theoretically deduced deviations (Ref. 25) due to interionic friction amount to some 30% at $T > T_0$, so any salt with $\alpha \geqslant 0.7$ by this analysis should be considered to be fully dissociated.

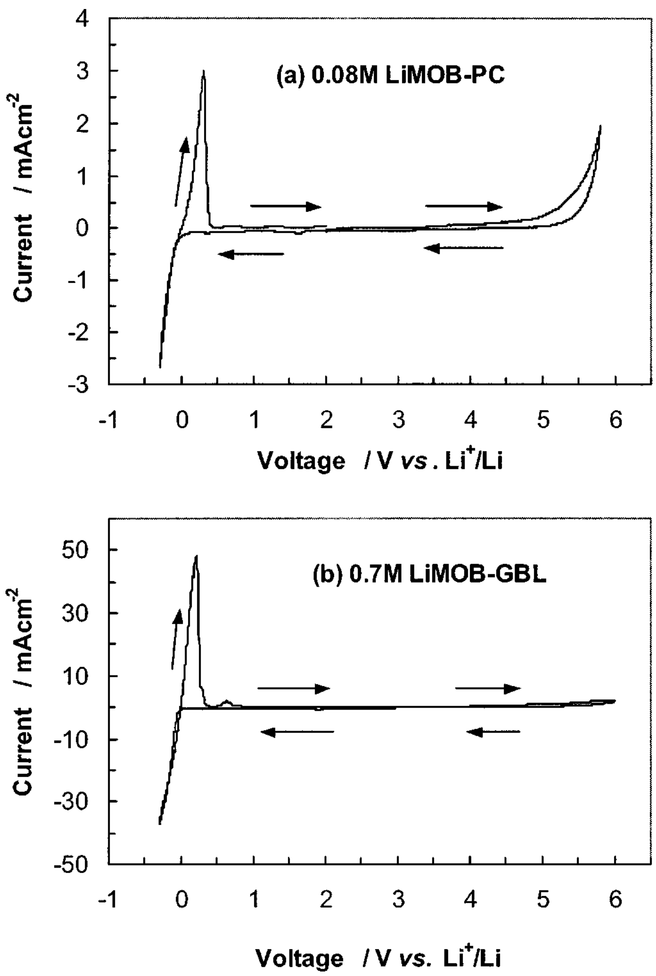


**Figure 6.** Cyclic voltammograms for LiMOB solutions in (a) PC and (b) GBL on platinum electrode (area $4.91 \times 10^{-4}$ cm$^2$) with a scan rate of 5 mV s$^{-1}$ at room temperature.

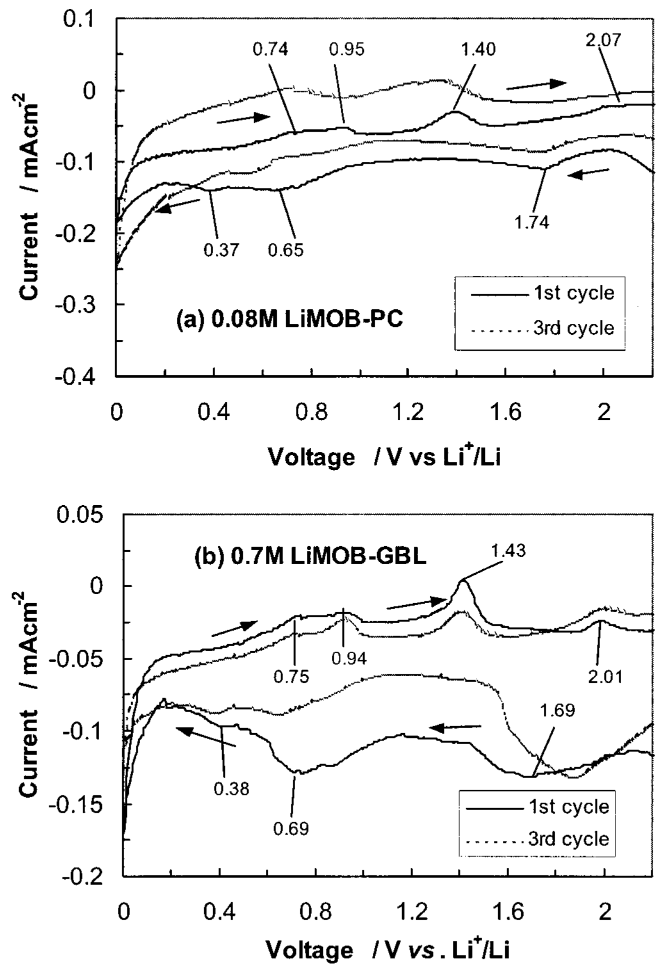


**Figure 7.** Cyclic voltammograms of LiMOB solutions in (a) PC and (b) GBL on platinum electrode (area $1.96 \times 10^{-3}$ cm$^2$) in the voltage range of $0 \sim 2.2$ V, with a scan rate of 5 mV s$^{-1}$ at room temperature.

reduction of trace water and adsorbed to the platinum electrodes.

Several studies have shown that measured oxidation potentials ($E_{ox}$) can be correlated with calculated anion HOMO energies. Yilmaz *et al.*[23] found a linear correlation between $E_{ox}$ of unsaturated organic molecules and the HOMO energies calculated by the modified neglect of diatomic overlap (MNDO) method. Barthel *et al.*[4] calculated MNDO and AM1 HOMO energies of fluorinated bis[1,2-benzenediolato(2-)-O,O′]borates $[B(C_6H_{4-x}F_xO_2)_2]^-$. The two sets of HOMO energies were similar, and both were correlated with $E_{ox}$ (as defined by the anodic oxidation limit of the lithium borates, $Li[B(C_6H_{4-x}F_xO_2)_2]$). Increasing fluorination ($x$) of the anion increased $E_{ox}$ by $\sim$0.1 V/$x$ and decreased the HOMO energies by $\sim$0.3 eV/$x$.

Several studies of anion oxidation potentials were recently updated and extended by Ue *et al.*[24] They performed two types of calculations on the anions, obtaining HOMO energies from *ab initio* Hartree-Fock calculations and adiabatic ionization potential ($I_p$) from hybrid density functional B3LYP calculations. They found that $E_{ox}$ was correlated with HOMO energies and with $I_p$. They also repeated the calculations with three different basis sets, 6-31G*, 6-31 + G**, and 6-31 + G(2d,p), and found the latter two "diffuse" basis sets gave essentially identical energies.

Here we report HOMO energies of different orthoborate anions calculated using Hartree-Fock/6-31 + G** and B3LYP/6-31 + G** methods (Table II). The (difluoromalonato oxalato)borate anion, bis(difluoromalonato)borate anion, and bis(perfluoropinacolato)borate anion, are abbreviated DFMOB$^-$, BDFMB$^-$, and BPFPB$^-$, respectively. These data show that the magnitudes of the HOMO energies are sensitive to the computational method, but structural trends in HOMO energy are not. The two sets of energies are strongly correlated ($r = 0.999$) and both span narrow ranges (0.94 and 0.78 eV for HF and B3LYP energies, respectively). The data fail to reveal a significant difference between anions containing oxalate and malonate ligands. Fluorine substitution, on the other hand, lowers the HOMO energy. If HOMO energy is correlated with

**Table II. HF/6-31 + G** and B3LYP/6-31 + G** HOMO energies of orthoborate anions (B3LYP/6-31 + G** equilibrium geometries).**

| Anion | $E_{HOMO}$ (HF) (eV) | $E_{HOMO}$ (B3LYP) (eV) |
|---|---|---|
| BMB$^-$ | −8.73 | −4.27 |
| MOB$^-$ | −8.81 | −4.34 |
| BOB$^-$ | −8.87 | −4.39 |
| DFMOB$^-$ | −9.03 | −4.55 |
| BDFMB$^-$ | −9.56 | −5.00 |
| BPFPB$^-$ | −9.67 | −5.05 |

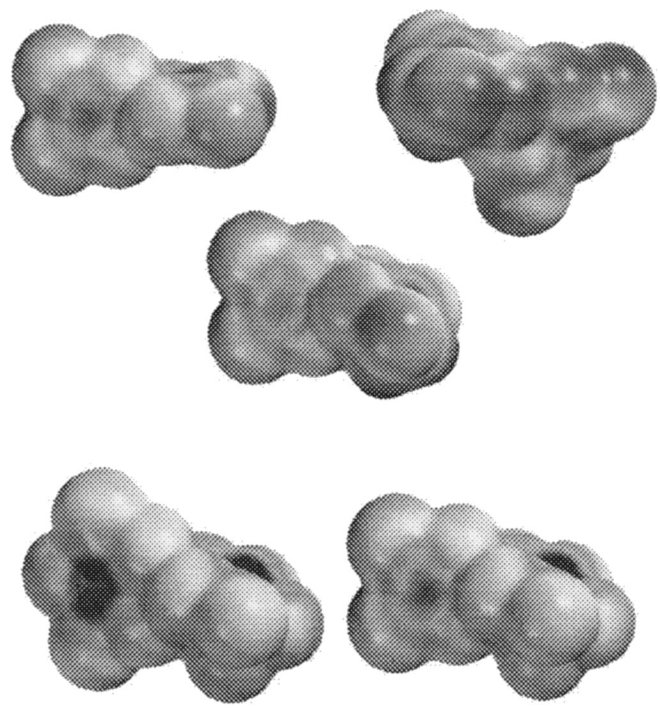

**Figure 8.** Electrostatic potential maps of all three anions ($BOB^-$ upper left, $BMB^-$ upper right, $MOB^-$ central, $BDFMB^-$ lower left, and $DFMOB^-$ lower right). (Color maps are available from the corresponding author on request.)

$E_{ox}$, as expected, the $DFMOB^-$ and $BDFMB^-$ anions are predicted to have higher electrochemical oxidation potentials than their corresponding nonfluorinated orthoborate anions ($MOB^-$ and $BMB^-$). Thus fluorination of the -$CH_2$- groups could improve the electrochemical oxidation stability. It should also improve solubility, and possibly also ionic mobility.

*Computational results: Structure and electrostatic maps*.—The electronic character of the asymmetric orthoborate anion, $MOB^-$, is depicted by its electrostatic potential map in Fig. 8, along with those of $DFMOB^-$ and $BDFMB^-$, and of $BOB^-$ and $BMB^-$ that have been reported in Ref. 14 (color maps are available from the corresponding author upon request).

Because these anions are relatively large delocalized species, the map potentials span a relatively narrow range: red ($\leqslant -106$ kcal/mol) < orange < yellow ($-89.5$ kcal/mol) < green ($-73$ kcal/mol) < light blue ($-56.5$ kcal/mol) < dark blue ($\geqslant -40$ kcal/mol). The most negative potentials (red and orange) are found around, and in between, the terminal oxygens of the ligand carbonyl (C = O) groups. The most negative potential is relatively constant from one anion to the next (Table III). More variation is seen at the other extreme, that is, in the potentials found near ''neutral'' atoms. This makes sense since there is considerably more structural variation in the neutral parts of these ligands.

The maps suggest that, to some degree, each ligand establishes its electron distribution independently of the other ligand, e.g.

1. The three anions that contain oxalyl ligands ($BOB^-$, $MOB^-$ and $DFMOB^-$) consistently create about the same negative potential ($-106$ to $-110$ kcal/mol) between the carbonyl oxygens of the oxalyl ligand.

2. The two anions ($BMB^-$ and $MOB^-$) that contain a malonyl ligand create about the same potential ($-36$ to $-39$ kcal/mol) near the $CH_2$ hydrogens.

3. The two anions ($DFMOB^-$ and $BDFMB^-$) that contain a difluoromalonyl ligand create about the same potential ($-22$ to $-25$ kcal/mol) over the map face that lies closest to the three carbon atoms in the ligand.

The optimized geometries of the three orthoborate anions are also listed in Table III. Note that the italicized O-B-O bond angles are between oxygens in the same bidentate ligand.

## Conclusions

A new lithium salt, with asymmetric chelatoborate anion based on two different dicarboxylates, has been synthesized. The salt, LiMOB, is thermally stable but unfortunately only weakly soluble in many of the common organic solvents used in batteries. However, the solubility of LiMOB in certain solvents such as DMSO and DMF is quite high, and the solutions are highly conductive and electrochemically stable. The conductivities of LiMOB solutions are close to those of LiBOB solutions so LiMOB may also prove to be a successful replacement for $LiPF_6$ in lithium-ion batteries. A correlation between the HOMO energy and the electrochemical stability, and the relation between the electronic structures and physical properties of the new and other weakly coordinating orthoborate anions, including their fluorinated derivatives, both suggest that a similar salt with difluorination of the methylene group would be more soluble than the unfluorinated salt, and could have unique properties. Work on such fluorinated salts is in progress.

**Table III. Extreme potentials and optimized geometry parameters of orthoborate anions.[a]**

| Anion (point group) | Least negative potential (kcal/mol) | Most negative potential (kcal/mol) | B-O bond distance (Å) | O-B-O bond angle (°) | |
|---|---|---|---|---|---|
| | | | | Intrabidentate ligand | Interbidentate ligand |
| $MOB^-$ ($C_s$) | $-36$ (near C$\boldsymbol{H}_2$) | $-110$ (between ox C = O) | 1.467 (ox), 1.501 (ox), 1.466 (mal) | *105.2* (ox), *112.2* (mal) | 109.7, 109.9 |
| $BOB^-$ ($C_2$) | $-43$ (over ox ***C***) | $-107$ (between ox C = O) | 1.476 | *105.6* | 111.4 |
| $BMB^-$ ($C_s$) | $-39$ (near C$\boldsymbol{H}_2$) | $-109$ (on ridge created by two $CO_2$) | 1.458, 1.459, 1.490, 1.490 | *111.9*, *111.9* | 108.1, 108.1, 108.3, 108.4 |
| $DFMOB^-$ ($C_s$) | $-25$ (over dfm ***C***) | $-106$ (between ox C = O) | 1.465 (ox), 1.485 (ox), 1.472 (dfm) | *105.9* (ox), *112.4* (dfm) | 109.1, 110.1 |
| $BDFMB^-$ ($C_2$) | $-22$ (over dfm ***C***) | $-97$ (between $CO_2$) | 1.463, 1.480 | *112.7* | 106.9, 107.7, 107.7, 109.2 |

[a] ox = oxalyl, mal = malonyl, dfm = difluoromalonyl.

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